

PERSONAL INFORMATION

Name: Erjia (Eric) Guan
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EDUCATION

MASc Master of Applied Science in Civil Engineering, University of Toronto, Canada | 2024 - 2026

Department: Civil & Mineral Engineering

Supervisors: Prof. Heather MacLean and Prof. Daniel Posen

Thesis Title: Resource Intensity and Deployment Barriers of Direct Air Capture at Scale in Canadian Net-Zero Pathways

Project: Assessed technological, economic, institutional, geophysical, environmental-ecological and socio-cultural feasibility using the IPCC AR6 WGIII framework. Quantified energy, water, and land requirements across Canadian net-zero deployment scenarios through 2050.

BScH Bachelor of Science (Honours) in Environmental Life Sciences, Queen's University, Canada | 2019 - 2024

Undergraduate thesis: An Assessment of Phosphorus Fractions in the Canadian High Arctic Post-Permafrost Disturbances

Supervisor: Prof. Neal Scott

Project: Investigated high arctic soil phosphorus fractions using a chronosequence of three permafrost disturbances spanning 1-70 years and paired controls. Applied Hedley sequential fractionation and spectrophotometric analysis to quantify phosphorus pools and assess variation with time from disturbance.

RELEVANT EXPERIENCE

Graduate Researcher University of Toronto, Toronto, ON

May 2025 - Sep 2026

Focus: Life Cycle Assessment of Reusable and Single-Use Plastics

Contributed to ExxonMobil-funded research comparing reusable and single-use plastics through life cycle assessment, literature synthesis and data-quality analysis. Evaluated evidence from published studies to examine how data quality and methodological choices affect environmental comparisons.

Focus: Resource Requirements and Feasibility of Direct Air Capture in Canadian Net-Zero Pathways

Led parts of CANC02Re Activity 13, assessing direct air capture feasibility in Canada. Synthesized resource-intensity estimates from LCA and laboratory studies and applied them to NAVIUS net-zero scenarios with non-zero DAC deployment.

Research Assistant

May 2022 - Aug 2023

Department of Geography and Planning, Queen's University

Investigated soil organic matter distribution and quality in High Arctic semi-deserts and phosphorus fractions following permafrost disturbances. Conducted repeated gas sampling, chromatographic measurements and soil phosphorus analyses.

Teaching Assistant
Sep 2025 - Jan 2026

CIV300 - Terrestrial Energy Systems, University of Toronto, St George
Led biweekly in-person tutorials for approximately 40-50 students, explaining course concepts and supporting problem-solving. Graded midterm assessments, tests and the final examination. Held office hours to facilitate 1-on-1 teaching.

TECHNICAL SKILLS AND RESEARCH METHODS

- Life cycle assessment and inventory analysis: Familiarity conducting life cycle assessments using openLCA, SimaPro. Moderately familiar with Brightway. Experience creating life cycle inventories using the GaBi and ecoinvent databases. Experience evaluating data quality, system boundaries and assumptions when comparing products and processes.
- Material flow analysis: Quantified material inputs, outputs and stocks; constructed material balances and assessed resource use and flows across product systems and supply chains.
- Uncertainty and sensitivity analysis: Applied Monte Carlo simulation, scenario analysis and sensitivity testing to evaluate uncertainty, identified influential parameters and assessed the robustness of model results.
- Feasibility and environmental impact assessment: Assessed environmental impacts, resource requirements and implementation barriers.
- Quantitative decision modelling: Built quantitative decision-making models in Microsoft Excel using PrecisionTree and related decision-analysis software; compared alternatives using decision trees and expected values
- Markov chain analysis: Developed and analyzed state-transition models to assess how systems evolve over time and compare outcomes under alternative assumptions.
- Programming and data analysis: Familiarity with Python and R for data analysis.
- Microsoft Office

Laboratory Skills and Training

- Soil research: Hedley phosphorus fractionation; spectrophotometric determination of phosphorus concentrations; method development and reagent preparation using stoichiometry.
- Greenhouse gas measurements: Gas sampling from three sites over a 90-day period and gas chromatography to quantify changes in greenhouse gas concentrations.

CONFERENCES & PUBLICATIONS

Guan, E. (2023). An assessment of phosphorus fractions in the Canadian High Arctic post-permafrost disturbances. *Inquiry@Queen's Undergraduate Research Conference Proceedings*, 17. <https://doi.org/10.24908/iqurcp16351>

Guan, E. Posen, D. MacLean, H. (2026). *Resource intensity and deployment barriers of direct air capture at scale in Canadian net-zero pathways* [Master's thesis in preparation, University of Toronto].

Guan, E. (2025, June). *Resource Demands of Direct Air Capture at Scale, Canada* [Conference presentation]. CANC02Re Conference, University of Calgary, Calgary, AB, Canada.

AWARDS, GRANTS, AND FELLOWSHIPS

USSRF Fellowship — Queen's University (\$9,900) | 2023

Awarded to support undergraduate research on high arctic phosphorus fractions.

Graduate Research Grant — University of Toronto (\$40,000) | 2024

Awarded to support research on life cycle of single use vs. alternative plastics and deployment feasibility of carbon dioxide removal technologies in Canada.

Ontario Graduate Scholarship (OGS) (\$20,000) | 2025

Awarded in recognition of academic excellence in graduate studies.

Conference Grant — University of Toronto (\$1,400) | 2025

Awarded to support conference participation at the University of Calgary and Kananaskis County.